

Data inspection

GDP per capita dataset: 193 rows × 303 columns

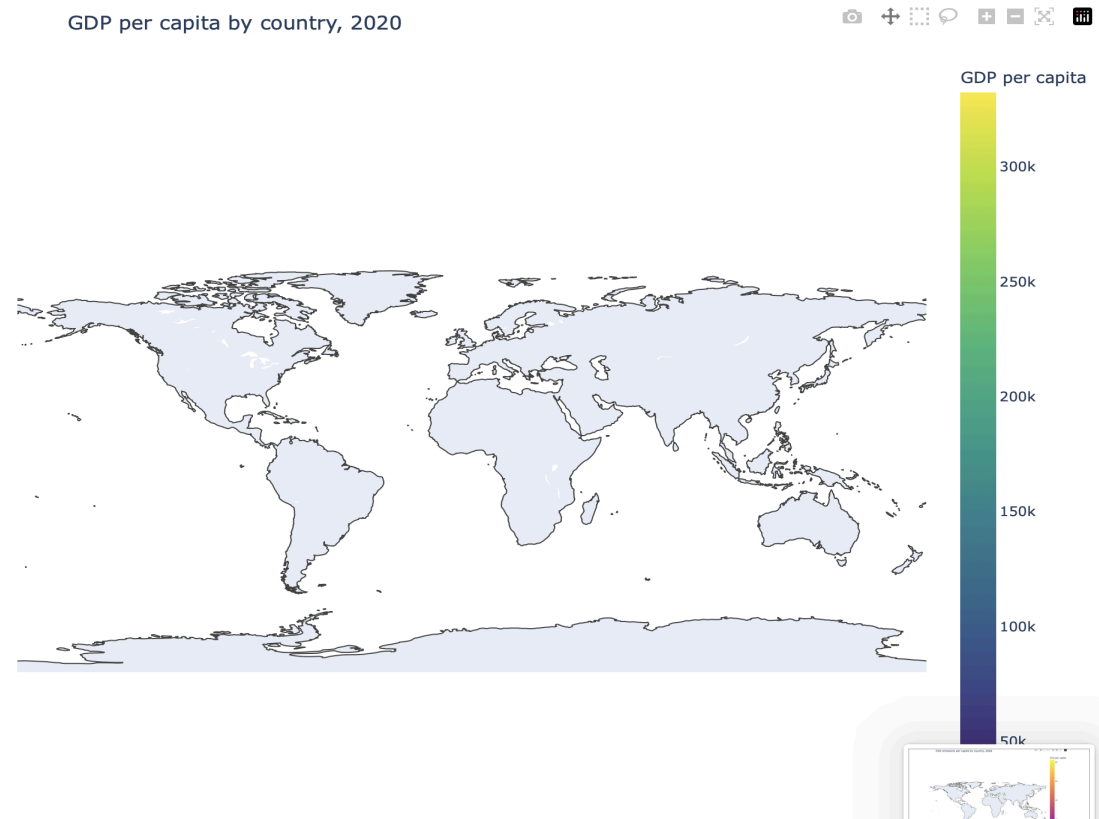
CO2 emissions per capita dataset: 192 rows × 225 columns

Both are in wide format: one row per country, with years as columns.

Dataset	Row s	Unique country codes	Duplicate country rows	Missing 2020 values
GDP per capita	193	193	0	0
CO2 per capita	192	192	0	0

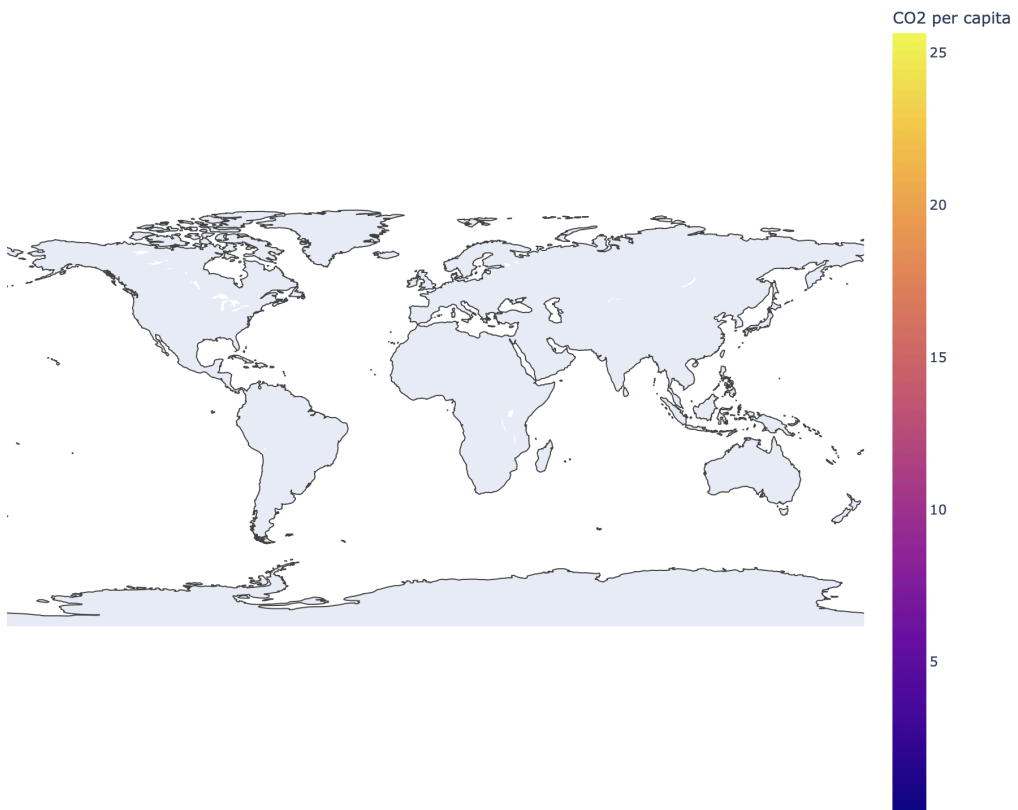
So the 2020 data is correctly structured for mapping: **one row per country**.

Outputs



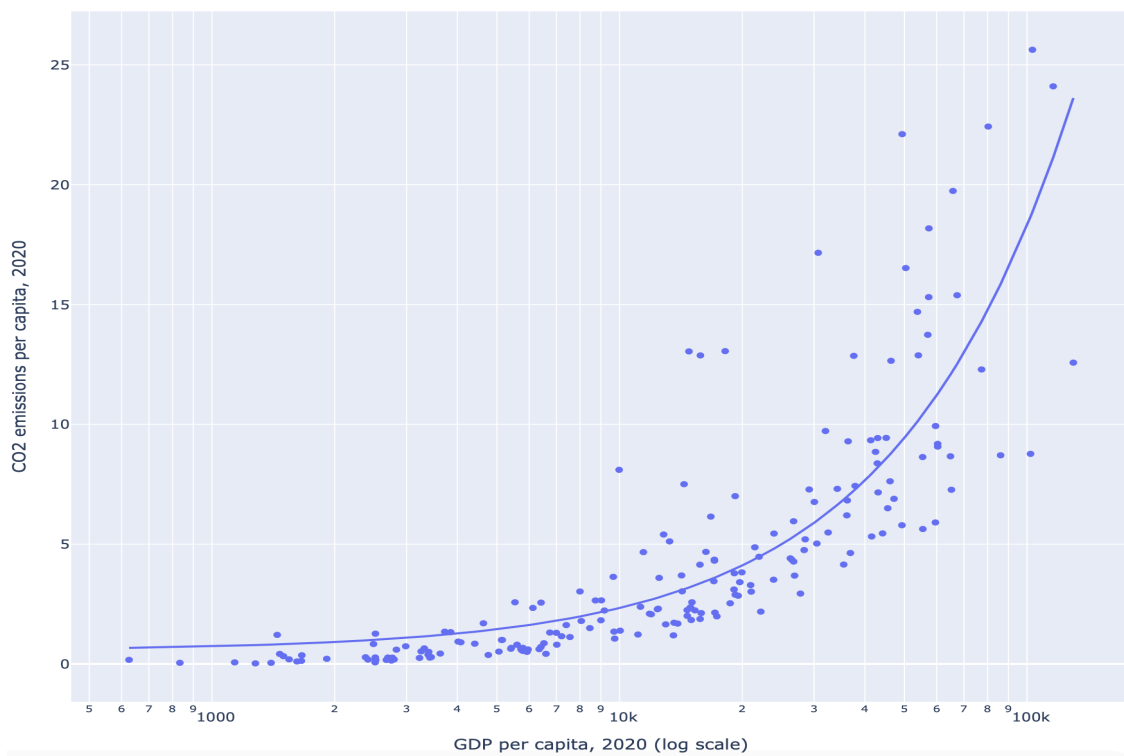
CO2 emissions per capita by country, 2020

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Relationship between GDP per capita and CO2 emissions per capita, 2020

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Brief interpretation

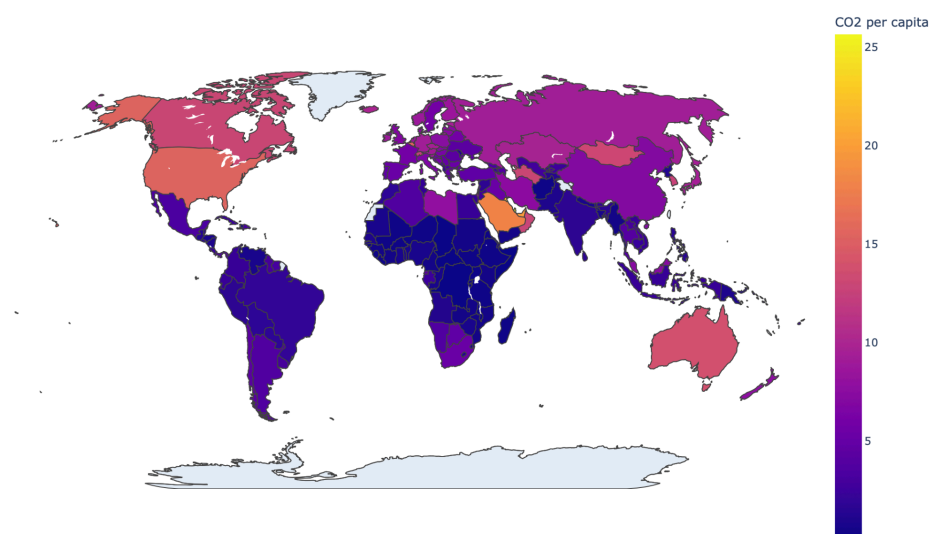
The scatterplot shows a **clear positive relationship** between GDP per capita and CO2 emissions per capita in 2020. Countries with higher GDP per capita generally tend to have higher CO2 emissions per person.

The relationship is fairly strong: the Pearson correlation is about **0.82** using GDP per capita directly, and about **0.75** when GDP per capita is log-transformed. This suggests that wealthier countries usually consume more energy and carbon-intensive goods and services per person, although the relationship is not perfect. Some high-income countries have lower emissions than others, likely due to differences in energy mix, industrial structure, climate, transport systems, and environmental policy.

Personal notes:

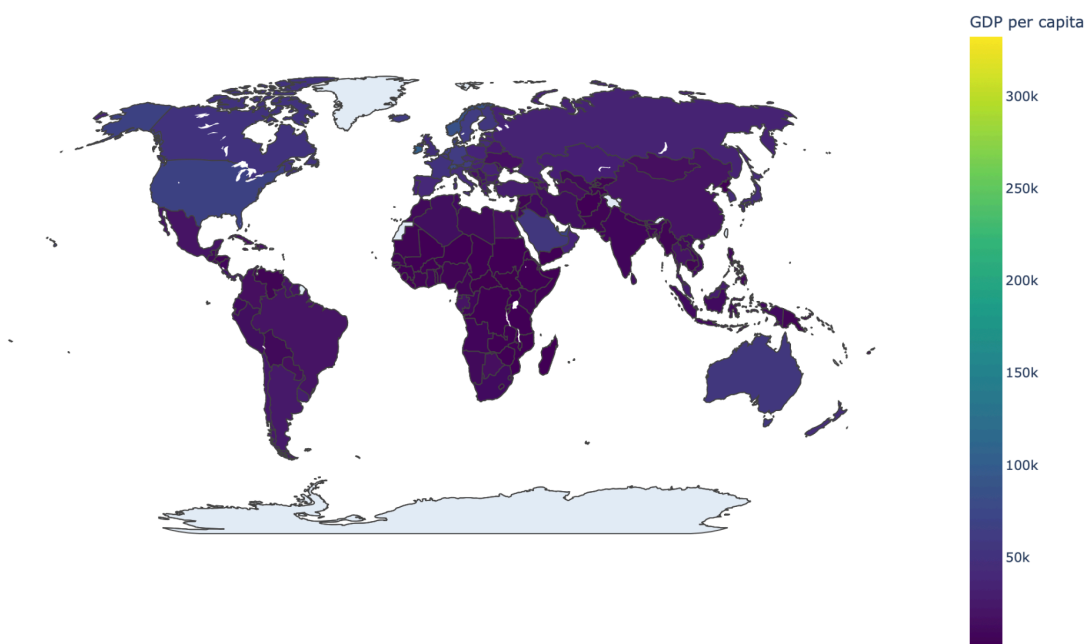
- the process seems quite easy: download the information and put it in an AI tool, here ChatGPT, to process the information and generate images that will help you understand the data and visualize it.
- but the colors don't appear on the maps... and instead of trying to fix it afterwards, I would like to achieve the "right outcome" from the first prompt! Here I added this after getting the first result: "the two maps dont have the colors integrated, fix this". This was the result:

CO2 emissions per capita by country, 2020



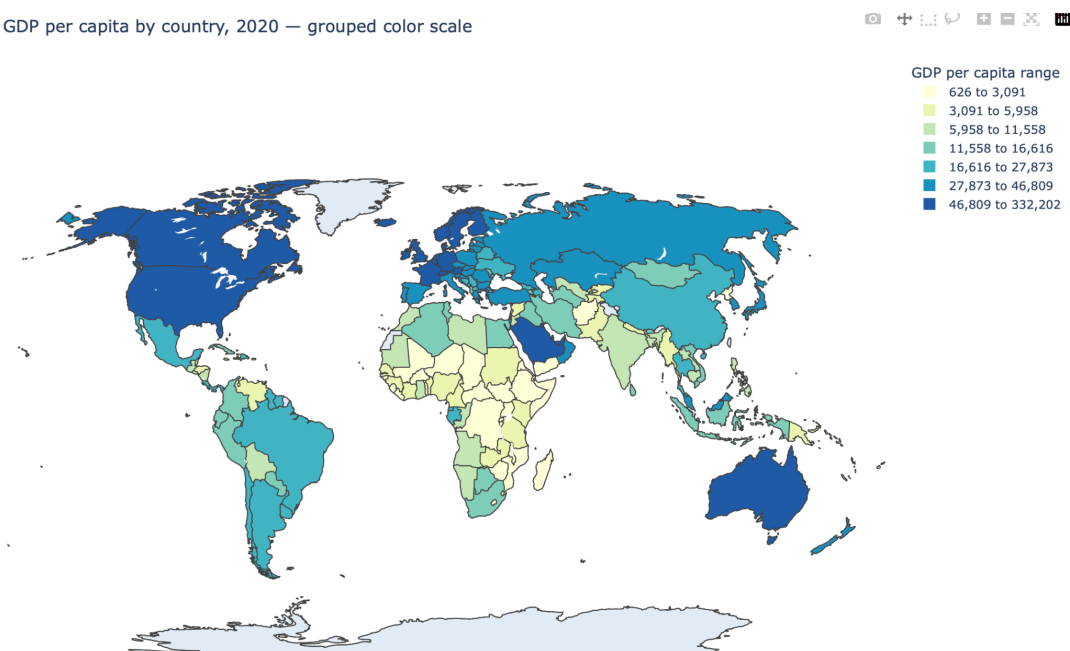
- but the second map wasnt good, because the colors were too similar:

GDP per capita by country, 2020



- so I added this request: "Fix the colors for the GDP per capita so I can see the difference between country. Right now the shades are too close to compare properly." For this result:

GDP per capita by country, 2020 — grouped color scale



Hopefully it is correct :)